

PROJECT PROPOSAL

Course: DATA-441: Applied Natural Language Processing

1. Title: Identifying Linguistic Drivers of Perceived Professor Difficulty: An Explainable Deep Learning Approach to Rate My Professor Reviews at American University.

2. Project Category This is an Application project, as it applies learning algorithms to solve a specific NLP task regarding text classification and sentiment analysis.

3. Team Details

- Jack Markert: DATA-441

4. Introduction/Motivation: The core problem I plan to tackle with this project is the lack of context behind quantitative education professor ratings. Students often heavily utilize Rate My Professor (RMP) ratings to navigate and help them make decisions during their course selections, relying on peer evaluations to gauge instructor difficulty and course quality. However, a quantitative "Level of Difficulty" rating (scaled 1.0 to 5.0) often provides little insight. Why might this professor be rated a "4" difficulty? Could it be from being a "tough grader", assigning "lots of readings", or giving "unclear lectures".

This problem is extremely relevant in the context of NLP because the true nuances of a student's experiences are locked within the unstructured and messy text of their review. By analyzing a localized dataset of roughly 34,000 Rate My Professor reviews from American University, I plan to extract the latent sentiment and specific linguistic triggers that drive these numerical difficulty ratings.

Additionally, as NLP heavily shift towards complex deep learning architectures, I feel the field faces a crisis of interpretability. Neural networks excel at text classification but operate as "black boxes." I plan for this project to address the critical need for Explainable AI (XAI) in educational data mining by not only predicting the difficulty rating but isolating the exact words and phrases that justify the model's prediction.

5. Methodology: This project will apply and compare machine learning techniques across the complete NLP pipeline. I plan to contrast highly interpretable traditional models with complex deep learning architectures to predict the "Level of Difficulty" rating from the text of roughly 34,000 student reviews at American University. Additionally, I plan to find and extract the specific words and triggers that drive students to give those difficulty rating.

- **Text Preprocessing:** The raw text will undergo rigorous preprocessing to standardize the vocabulary. This includes lowercasing, HTML tag removal, tokenization, filtering out standard stop-words, and applying lemmatization.
- **Interpretable Baseline Modeling:** I will first generate Term Frequency-Inverse Document Frequency (TF-IDF) vectors to capture word importance. These vectors will be fed into a Logistic Regression classifier. The linear nature of this model provides immediate explainability, allowing me to extract the exact coefficient weights of individual words that inherently drive high or low difficulty ratings.
- **Deep Learning & Contextual Modeling:** To capture multi-word phrases (e.g., "tough but fair") and long-term sequential dependencies inherent in lengthy student reviews, I will train Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). These neural networks will utilize Word2Vec embeddings (experimenting with both CBOV and Skip-Gram architectures) to capture deeper semantic relationships within the text.

6. Intended Experiments: My core experiment is to evaluate the trade-off between model accuracy and model explainability when classifying text.

- **Performance Evaluation:** The dataset of roughly 34,000 reviews will be split into training (80%) and testing (20%) sets. All models (Logistic Regression, CNN, RNN) will be evaluated on their ability to accurately predict the multi-class "Level of Difficulty" rating. I will measure the performance and effectiveness of the approach using standard classification metrics: Accuracy, Precision, Recall, and F1-Score.
- **Explainability Analysis (Unlocking the Black Box):** To satisfy the critical need for interpretability in deep learning, I will apply an explainability framework, specifically Local Interpretable Model-agnostic Explanations (LIME) or SHapley Additive exPlanations (SHAP), to the best-performing neural network. This will allow me to highlight the specific text regions (n-grams) the CNN or RNN focuses on when predicting a high or low difficulty rating.
- **Outcome Extraction:** The final deliverable will not just be a predictive model, but a tangible dictionary of the top linguistic drivers of student-perceived difficulty at American University, cross-verified by both the transparent baseline and the explainable neural networks.

7. Prior Research: Recent academic research validates the large power of subjective language in professor evaluations. In a recent working paper, Fernandez and Yetter conducted a comprehensive sentiment analysis on nearly 8 million Rate My Professor reviews. Their findings demonstrated a strong correlation between written sentiment and numerical evaluations, highlighting how quantitative ratings are heavily influenced by the subjective, unstructured text provided by students. My project builds upon this foundation by applying advanced deep learning and explainability frameworks to a localized university dataset to predict the specific difficulty metric.

8. References

Fernandez, J. M., and E. Yetter. 2025. "Sentiment Analysis of Teaching Evaluations: Evidence From Nearly 8 Million Rate My Professor Reviews." *Southern Economic Journal* 92, no. 2: 637–661. <https://doi.org/10.1002/soej.12787>.